
The Role of Scaling Symmetries in Linear and Nonlinear Mode Connectivity of Neural Networks

Anonymous Author
Anonymous Institution

Abstract

To analyze the loss landscape of neural networks, one has to take the weight symmetries into account. Neurons can be permuted without changing the function represented by the network. Therefore, when assessing whether two networks are similar, one must account for these symmetries. If, even after reparametrization or re-basing, interpolated networks show high loss (i.e. a high barrier), this may indicate that they lie in separate basins of the loss landscape and represent fundamentally different solutions. However, it has been conjectured that, for independently trained neural networks, some permutation re-basing removes such barriers entirely. We test this claim in ReLU networks using exhaustive experiments where permutation-search error can be ruled out. We find in small networks trained on the XOR problem perfectly accurate solutions that are fundamentally in different basins, providing a concrete counterexample to a universal permutation-based account of linear mode connectivity. We then enlarge the symmetry class with function-preserving positive scaling. We introduce both a joint Sinkhorn permutation-and-scaling method and a method-agnostic refinement that holds a chosen permutation fixed and optimizes only its weight scales. In these small networks, for which all permutations can be enumerated, scaling improves even the best permutation; across VGG/CIFAR-10 and MLP/Fashion-MNIST, it consistently reduces both weight-matching and Sinkhorn training-loss barriers. Nonlinear path experiments further separate weight symmetry from

architectural redundancy: the same small XOR networks stay separated even when non-linear Bézier and polygonal loss paths are used, whereas adding hidden units creates low-loss paths and makes weight scaling useful for path optimization. These results show that permutation symmetry is an incomplete explanation of the connectivity of neural-network loss basins and that both the optimization of functionally equivalent model representatives and the available model capacity determine whether low-loss paths are found.

1 INTRODUCTION

Are independently trained neural networks genuinely in different loss basins, or do they only appear separated because their hidden features use different coordinates? The question matters both for the geometry of neural-network loss landscapes and for operations such as weight averaging. A high-loss straight line between two parameter vectors is not conclusive: neural-network parameterizations are non-identifiable, and functionally equivalent representatives can have very different interpolation behavior.

The single-basin-modulo-permutation conjecture proposes that most stochastic gradient descent (SGD) solutions can be permuted into linear mode connectivity (LMC) (Entezari et al., 2021). Re-basing methods make this hypothesis operational and substantially reduce barriers in MLPs and convolutional networks (Ainsworth et al., 2023; Peña et al., 2023). Yet a residual barrier after alignment is ambiguous. An algorithm may have missed a suitable permutation, the permutation symmetry class may itself be too small, or the architecture may lack the representational freedom needed for a low-loss path. Large-network studies cannot cleanly distinguish these explanations because their permutation groups are combinatorial and alignment is approximate.

We isolate these mechanisms using small networks trained on the XOR problem, whose size permits exhaustive permutation search, then test the resulting account in VGG networks. The minimal-width experiment first removes the search confound: some trained pairs remain classification-incompatible under every permutation. To check that this behavior is not specific to deterministic full-batch optimization, we observe it with both full-batch gradient descent (GD) and minibatch SGD. We then enlarge the exact symmetry class using positive ReLU scaling. Starting from the exhaustive best permutation and optimizing scales alone identifies gains that cannot be attributed to better correspondence search. Finally, nonlinear path and width experiments separate representative choice from architectural redundancy: symmetry moves within an equivalence class, while extra units add routes through parameter space. Our contributions are:

- We give an exhaustive counterexample to universal permutation-based LMC for trained networks. At minimal XOR width, every pair of trained solutions not already nearly identical up to a hidden-unit permutation fails classification-LMC under exhaustive alignment.
- We introduce scale-aware alignment and show, after exhaustive permutation search, that positive scaling lowers the loss barrier for all tested pairs of networks trained on XOR at widths 3, 5, and 7.
- We show an empirical transition in nonlinear connectivity between widths two and three, distinguishing representational redundancy from symmetry alignment. Scaling both endpoints does not overcome the width-2 failure, but lowers the mean barrier at every tested width above two for both polygonal and Bézier parameterizations.
- We show that scaling improves over permutation-only alignment on three independently trained pairs for each of VGG11/13/16/19 on CIFAR-10 and a ten-layer MLP on Fashion-MNIST.

2 PROBLEM FORMULATION AND RELATED WORK

Connectivity and interpolation barriers. For endpoints $\theta_A, \theta_B \in \Theta$, linear interpolation is $\theta(\lambda) = (1 - \lambda)\theta_A + \lambda\theta_B$, with $\lambda \in [0, 1]$. Following Entezari et al. (2021), the barrier is

$$B(\theta_A, \theta_B) = \sup_{\lambda \in [0, 1]} \left[C((1 - \lambda)\theta_A + \lambda\theta_B) - [(1 - \lambda)C(\theta_A) + \lambda C(\theta_B)] \right]. \quad (1)$$

Here $C : \Theta \rightarrow \mathbb{R}$ is a scalar cost assigned to a model. For gradient-based alignment, C must be differentiable and denotes binary or multiclass cross-entropy in our experiments. We use C to follow the re-basing literature and to distinguish the generic cost from a particular choice of loss. Classification error is used only as a reporting cost and is never optimized. The supremum is approximated on a finite grid. Nonlinear mode connectivity instead asks whether a general curve $\gamma : [0, 1] \rightarrow \Theta$ joins the endpoints at low loss. A curved path may avoid a region crossed by the straight segment, so nonlinear connectivity does not imply LMC (Draxler et al., 2018; Garipov et al., 2018).

Permutation re-basing. Relabeling a hidden layer and applying the inverse relabeling to the next layer preserves the network function; the same action reorders channels in convolutional networks. Weight matching finds layerwise permutations that reduce Euclidean distance, while activation matching aligns representations on data (Ainsworth et al., 2023). Sinkhorn re-basing instead relaxes permutation matrices to differentiable, approximately doubly stochastic assignments that can be optimized against an interpolation objective (Peña et al., 2023). These successes support a symmetry-aware view of loss landscapes, but a residual barrier obtained by a heuristic cannot establish that the symmetry class itself is insufficient.

Why linear connectivity matters. A curved connection establishes geometric connectedness, but a straight path is the relevant object for parameter averaging. Checkpoint averaging can improve generalization within one trajectory (Izmailov et al., 2019), and averaging related fine-tuned models can be effective without ensemble-time cost (Wortsman et al., 2022). Our focus is the geometric prerequisite of a low-barrier segment, rather than a claim that an aligned midpoint must outperform its endpoints.

3 SCALE-AWARE ALIGNMENT

Permutation action. For a network with L affine layers, let $h_\ell = \sigma(W_\ell h_{\ell-1} + b_\ell)$ for hidden layers, where σ acts coordinatewise, and let P_ℓ relabel hidden layer ℓ . The induced action on adjacent weights and biases is

$$\widetilde{W}_\ell = P_\ell W_\ell P_{\ell-1}^\top, \quad \widetilde{b}_\ell = P_\ell b_\ell. \quad (2)$$

Because $P_\ell^{-1} = P_\ell^\top$ and coordinatewise activations satisfy $\sigma(P_\ell z) = P_\ell \sigma(z)$, Eq. 2 preserves the network function.

Original Sinkhorn re-basing. Sinkhorn re-basing replaces each discrete P_ℓ with a differentiable soft assignment (Peña et al., 2023). An unconstrained

score matrix Z_ℓ is mapped to an approximately doubly stochastic matrix $S_\ell = \text{Sinkhorn}(Z_\ell/\tau)$, where the temperature τ controls its sharpness. It optimizes the layerwise scores using the relaxed action

$$\widetilde{W}_\ell^{\text{soft}} = S_\ell W_\ell S_{\ell-1}^\top, \quad \widetilde{b}_\ell^{\text{soft}} = S_\ell b_\ell. \quad (3)$$

Gradients through the Sinkhorn map are obtained by implicit differentiation. Since $S_\ell^\top$ is generally not S_ℓ^{-1} , the soft action need not preserve the endpoint function. Linear assignment projects each S_ℓ to a hard, function-preserving permutation for evaluation.

Scale-aware extension. Permutation is only one class of exact symmetry in neural networks. Another is the scaling symmetry induced by positive homogeneity: for ReLU, $\sigma(cz) = c\sigma(z)$ for $c > 0$, so the incoming weights and bias of a unit can be multiplied by c , provided that its outgoing weights are divided by c . Applying permutation and positive scaling together is therefore an exact symmetry, selecting a different parameter-space representative of the same endpoint function. REPAIR also uses channel-wise scaling and shifting, but applies it to interpolated activations and thereby changes the represented function (Jordan et al., 2023). Consequently, the intermediate networks obtained by linearly interpolating to that endpoint can have different losses, and hence a different barrier. This motivates studying connectivity under a richer symmetry class before attributing a residual barrier to genuinely distinct basins.

We extend Sinkhorn because it already expresses permutation search as a continuous, differentiable optimization, so continuous scale variables can be added and trained jointly under the same interpolation objective. This choice does not imply that other re-basing methods cannot incorporate scaling. Doing so in layerwise matching or straight-through methods would instead require modifying their matching objectives or coupling discrete permutation updates with continuous scale optimization, making the extension less direct.

For each hidden layer, we introduce log-scales u_ℓ and set $D_\ell = \text{diag}(e^{u_\ell})$. The exponential enforces the positivity required by ReLU homogeneity. Combining D_ℓ with a hard permutation gives the following function-preserving transformation:

$$\widetilde{W}_\ell = P_\ell D_\ell W_\ell D_{\ell-1}^{-1} P_{\ell-1}^\top, \quad \widetilde{b}_\ell = P_\ell D_\ell b_\ell. \quad (4)$$

Positive homogeneity propagates the scaled, permuted activation to the next layer, where the inverse factor cancels it; hence the transformed endpoint computes the same function.

During joint soft optimization, our implementation uses the corresponding relaxation

$$\widehat{W}_\ell = S_\ell D_\ell W_\ell D_{\ell-1}^{-1} S_{\ell-1}^\top, \quad \widehat{b}_\ell = S_\ell D_\ell b_\ell. \quad (5)$$

Only the diagonal scaling is inverted exactly; $S_{\ell-1}^\top$ remains the soft surrogate used by the original method. Projecting S_ℓ to P_ℓ recovers the exact action in Eq. 4.

We consider two variants. *Permutation + scale (joint Sinkhorn)* optimizes Z_ℓ and u_ℓ together. *Permutation + scale (fixed-permutation refinement)* first obtains and freezes a hard permutation, then optimizes u_ℓ alone. A quadratic penalty on u_ℓ controls the unbounded scaling symmetry and discourages numerically ill-conditioned representatives. Extending the standard re-basin notation $\pi_P(\theta_B)$, let $\pi_{Z,u}(\theta_B)$ denote the endpoint transformed by the current soft assignments and scales. Following the original Sinkhorn method (Peña et al., 2023), we use the interpolation cost

$$C_\lambda(Z, u; \theta_A, \theta_B) = C((1 - \lambda)\theta_A + \lambda\pi_{Z,u}(\theta_B)). \quad (6)$$

Fixing $\lambda = 0.5$ gives the midpoint objective; drawing $\lambda \sim \mathcal{U}(0, 1)$ independently at each update gives the random-point objective. The midpoint is inexpensive and often the most vulnerable location, whereas random sampling covers the path across updates. We minimize Eq. 6 plus $\lambda_s \sum_\ell \|u_\ell\|_2^2$, where the sum is over hidden layers. Freezing a hard permutation removes Z from the optimization. We report barriers only after projecting the soft assignments to hard permutations. Experiment-specific objectives and grids are given in Appendix A.

4 CONTROLLED XOR STUDIES

All studies in this section use one-hidden-layer ReLU networks with biases and one output logit, trained on the four XOR examples. Endpoints are initialized independently and retained only if they reach 100% classification accuracy and low binary cross-entropy. Unless stated otherwise, endpoint training uses full-batch GD, and connectivity is evaluated using the barrier in Eq. 1. This controlled setting lets us enumerate the complete hidden-unit permutation group at every studied width. Full training and optimization details appear in Appendix A.

4.1 Permutation Symmetry Is Insufficient

When permutation-based LMC fails, did the alignment algorithm fail, or does no permutation work? We first answer this in the smallest nontrivial setting, width $h = 2$. There are only $2! = 2$ hidden-unit permutations, so the symmetry class can be searched exactly.

Training and evaluation. As a robustness check against the interpretation that the effect is specific to deterministic optimization, we repeat the default full-batch protocol with minibatch SGD using minibatches containing two of the four XOR examples and

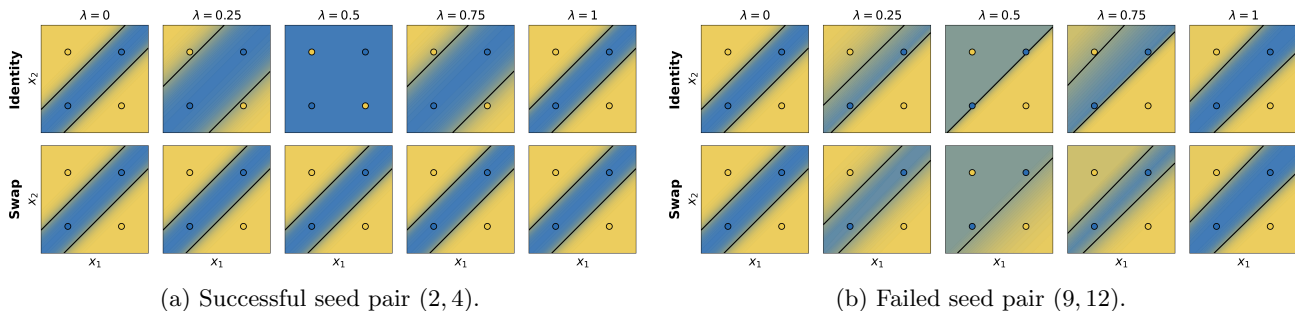


Figure 1: Decision boundaries along linear interpolation paths for two width-2 XOR endpoint pairs. Columns show interpolation values λ ; rows compare the identity and swap permutations. “Successful” means that at least one permutation preserves 100% accuracy along the entire segment; “failed” means that both permutations incur interpolation error despite perfectly accurate endpoints.

reshuffling each epoch. For each protocol, eight successful endpoints give 28 pairs. We enumerate identity and swap and evaluate classification error along each interpolation.

Result. For both training protocols, only 6/28 pairs preserve 100% accuracy along the best-permuted segment; the remaining 22/28 incur interpolation error under both possible permutations. Figure 1 shows a successful pair and a representative full-batch failure. In the failure, both endpoints correctly implement XOR and have similar decision boundaries, but no hidden-unit relabeling makes their straight interpolation compatible. Because the search is exhaustive, this result cannot be attributed to weight matching, Sinkhorn relaxation, or alignment initialization.

These trained endpoints refute a universal claim that permutation-based LMC follows automatically from SGD or GD. They do not contradict the sufficiently-wide, high-probability form of the single-basin-modulo-permutation conjecture. They also qualify the interpretation of the hand-constructed counterexample of Ainsworth et al. (2023). Ainsworth et al. used adversarial, non-SGD endpoints to show that architecture alone does not guarantee permutation-based LMC, and argued that the empirical prevalence of LMC may therefore reflect an implicit bias of SGD. Our exhaustive result shows that this bias is not absolute: standard minibatch SGD and full-batch GD can themselves produce successful solutions for which no permutation yields LMC.

4.2 Positive Scaling Beyond Permutation

Does positive scaling improve compatibility once permutation-search error has been removed? Using the shared full-batch protocol at widths $h = 3, 5, 7$ gives 21, 36, and 10 valid endpoint pairs. For each pair we enumerate all $h!$ permutations, select the one with

the smallest sampled loss barrier, and then freeze that permutation while optimizing only positive scales. Final barriers use 61 interpolation points. We also report raw interpolation and practical Sinkhorn baselines.

Width and permutation alignment. Increasing width changes the result. In the initial exhaustive sweep over $h \in \{2, 3, 4, 5\}$, the number of pairs maintaining 100% accuracy along the best-permuted interpolation rises steadily; at $h = 5$, every evaluated pair maintains perfect accuracy. Perfect accuracy does not mean zero loss barrier: midpoint confidence can still fall relative to the endpoints.

This width trend is consistent with Git Re-Basin, which empirically found lower post-alignment barriers for wider VGG and ResNet models (Ainsworth et al., 2023). Our purpose here is not to re-establish that trend, but to measure it under exhaustive permutation search and then isolate the additional effect of scaling. Table 1 reports all aggregate results. The exhaustive-permutation mean falls from 0.357 at $h = 3$ to 0.023 at $h = 7$; practical Sinkhorn remains above the exhaustive result at each width, so failure of the heuristic alone cannot establish that no suitable permutation exists.

Variability across pairs. Barriers are nonnegative and strongly right-skewed: some independently trained pairs are almost connected, whereas others remain more disconnected. For example, at $h = 5$ the joint Sinkhorn+scale barrier has median 0.011 but ranges up to 0.692, yielding 0.074 ± 0.149 . Pairs share endpoints, so these statistics are descriptive.

Scale beyond exhaustive permutation. The decisive comparison is between exhaustive permutation and the same frozen permutation followed by scale refinement. Refinement lowers the mean barrier by 31–78% across widths and lowers it on all 67 evaluated pairs. Since all permutations have already been

Alignment method	$h = 3$	$h = 5$	$h = 7$
No alignment	0.780 ± 0.610	0.608 ± 0.465	0.331 ± 0.184
Permutation only (Sinkhorn)	0.444 ± 0.325	0.189 ± 0.162	0.063 ± 0.044
Permutation only (exhaustive)	0.357 ± 0.309	0.111 ± 0.118	0.023 ± 0.015
Permutation + scale (joint Sinkhorn)	0.428 ± 0.380	0.074 ± 0.149	0.009 ± 0.009
Exhaustive permutation + scale refinement	0.247 ± 0.283	0.025 ± 0.044	0.005 ± 0.003

Table 1: Mean $\pm$ standard deviation of the XOR loss barrier across endpoint pairs, where h is the hidden-layer width. Bold indicates the lowest mean at each width; lower is better.

enumerated and the selected permutation is held fixed during refinement, the comparison isolates the additional rescaling freedom. For these endpoints, rescaling reaches functionally equivalent parameterizations with lower interpolation barriers than are attainable by permutation alone.

Joint optimization is more mixed at the smallest width. It achieves a lower mean barrier than exhaustive permutation at $h = 5$ and $h = 7$, but not at $h = 3$. Its gap to fixed-permutation scale refinement is therefore an optimization gap: the richer symmetry class contains better representatives even when a from-scratch solver does not reliably find them. Appendix Figure A.1 shows the corresponding mean interpolation profiles. The curves visualize the width trend and method ranking summarized more completely by Table 1.

4.3 Width Enables, Scaling Refines Nonlinear Connectivity

Residual linear barriers need not imply disconnected low-loss sets, so we test both the polygonal and quadratic Bézier paths used by Garipov et al. (2018). For endpoints θ_A, θ_B and one trainable interior control point v , their one-control-point parameterizations are

$$\phi_v^{\text{poly}}(\lambda) = \begin{cases} (1 - 2\lambda)\theta_A + 2\lambda v, & \lambda \in [0, \frac{1}{2}], \\ 2(1 - \lambda)v + (2\lambda - 1)\theta_B, & \lambda \in [\frac{1}{2}, 1], \end{cases} \quad (7)$$

$$\phi_v^{\text{Bézier}}(\lambda) = (1 - \lambda)^2\theta_A + 2\lambda(1 - \lambda)v + \lambda^2\theta_B. \quad (8)$$

The polygonal path consists of two line segments meeting at v ; the Bézier path bends smoothly under its influence. For the path-only searches, we keep the endpoints fixed and optimize the control points to minimize the maximum loss over 31 sampled path locations, using up to ten interior points. To test whether width affects available routes rather than only alignment, we apply these paths to endpoints from the shared protocol. At widths $h \in \{2, 3, 5, 7\}$, we also compare a one-control-point path with joint optimization of that path and independent positive scales at both raw endpoints. No permutation alignment is applied in this comparison. The transformed endpoints and joint optimization objective are given in Appendix A.2.

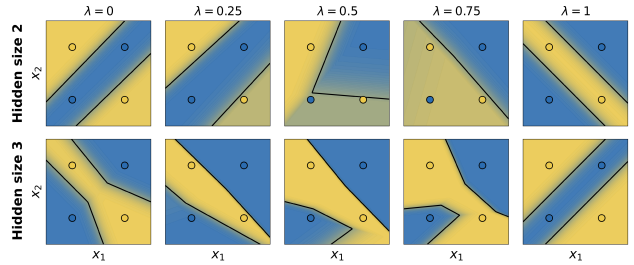


Figure 2: Representative quadratic Bézier paths for XOR. At width 2, optimizing the path does not remove the high-loss region; at width 3, a low-loss path connects endpoints with different decision boundaries. Columns correspond to different interpolation coefficients λ , and hence to different locations along the learned nonlinear interpolation path; colors denote predicted classes and markers show the four fixed XOR inputs. Only the network parameters vary across columns, and every displayed width-3 network classifies all four inputs correctly.

Result. At $h = 2$, searches find low-loss nonlinear paths only between essentially identical endpoints; additional interior points do not recover nontrivial connections. At $h = 3$, all studied endpoint pairs are connected by low-loss paths that retain 100% accuracy throughout interpolation. Figure 2 contrasts outcomes. This result is empirical, not a proof that no width-2 path exists outside the tested families. The only two successful width-2 endpoint pairs correspond to pairs (2, 9) and (5, 12). These pairs are nearly duplicate parameter solutions: their Euclidean endpoint separations are only about 0.6% of the corresponding parameter norms, and both path families attain barriers at numerical zero, providing no evidence of a nontrivial connection between distinct width-2 solutions.

A plausible explanation for the remaining failures is that width 2 is capacity-tight for XOR. The nontrivial endpoint pairs have different decision-boundary geometries, each constructed from only two ReLU boundary pieces. Within the tested path families, continuously transforming one geometry into the other with no spare hidden unit requires an intermediate arrangement that

misclassifies at least one XOR point. After adding a third unit, all studied pairs admit low-loss paths, consistent with the extra unit providing the temporary representational freedom needed during this transformation. This supports a distinction that endpoint alignment alone cannot make: symmetry selects among equivalent representatives at fixed capacity, whereas width changes the representational resources available along a path.

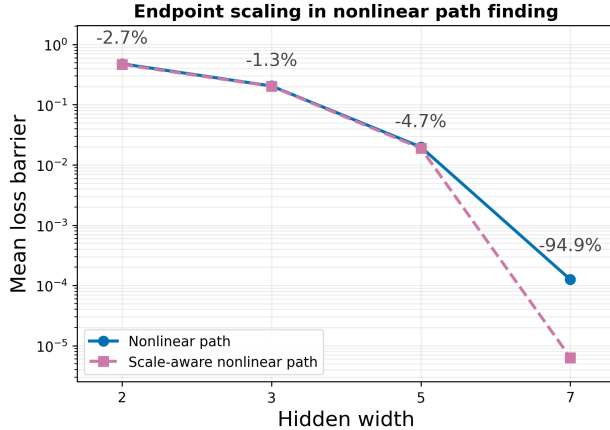


Figure 3: Mean nonlinear-path loss barrier by hidden width for polygonal paths with one interior control point, optimized from raw endpoints with or without independent positive scaling of both endpoints. Percentage annotations report the relative change under endpoint scaling; both curves aggregate the same endpoint pairs at each width.

Endpoint-scaling result. Endpoint scaling does not remove the minimal-width obstruction. With one interior control point, both polygonal and Bézier searches find low-loss connections for only 2/28 width-2 pairs, with or without jointly optimized endpoint scales. At larger widths, scaling nevertheless improves path finding (Figure 3), with quadratic Bézier paths behaving similarly. Geometrically, joint optimization fits the nonlinear curve while allowing each endpoint to move continuously along its loss-preserving symmetry orbit. Thus, lower-loss routes exist but may be easier to express or find after selecting suitable functionally equivalent endpoint representatives.

5 LARGER-NETWORK VALIDATION

Following the permutation-aligned connectivity benchmarks of Jordan et al. (2023), we evaluate VGG11, VGG13, VGG16, and VGG19 on CIFAR-10, and a ten-layer width-512 MLP on Fashion-MNIST. For every architecture–dataset combination, six independently

trained models form three disjoint endpoint pairs: runs initialized with seeds (0, 1), (2, 3), and (4, 5). As a data-independent permutation baseline, weight matching (WM) approximately solves the compact sum of bilinear assignments problem (SOBLAP)

$$(P_1^*, \dots, P_{L-1}^*) \in \arg \min_{P_1, \dots, P_{L-1}} \|\theta_A - \pi_P(\theta_B)\|_2^2, \quad (9)$$

where θ concatenates all weights and biases, and π_P applies the hidden-layer relabelings of Eq. 2. The standard solver cycles through layers, solving one linear assignment problem while holding the remaining permutations fixed (Ainsworth et al., 2023). For WM with scale refinement, we fix the resulting permutation and optimize only the function-preserving positive scales.

We compare raw interpolation, weight matching, weight matching followed by positive-scale refinement, Sinkhorn alignment, joint Sinkhorn–scale optimization, and positive-scale refinement of the projected Sinkhorn permutation. The complete hyperparameter-selection and evaluation protocol is given in Appendix A.4. Because the datasets and architectures differ, comparisons are made between methods within each model rather than between absolute barrier magnitudes across groups.

Result. Across all five architecture–dataset settings in Figure 4, positive-scale refinement reduces the mean WM barrier by 64–87%. It also consistently improves a fixed Sinkhorn permutation. Under the reported hyperparameter grid, WM followed by scale refinement outperforms joint Sinkhorn–scale optimization and performs best in every setting. Sinkhorn-based methods are more sensitive to hyperparameter selection, and broader exploratory searches produced better alignments in some settings. The comparison therefore characterizes performance under the standardized search protocol rather than establishing a definitive ranking of the alignment methods. None of the reported configurations eliminates the sampled training-loss barrier.

Across the studied architecture–dataset combinations, this extends the controlled XOR finding beyond small networks. The consistent gains after both WM and Sinkhorn alignment show that permutation and positive-scaling symmetries are complementary: permutation-aligned endpoints need not be the functionally equivalent representatives with the lowest linear barrier. The remaining barriers show that the tested searches improve, but do not recover, barrier-free linear connectivity.

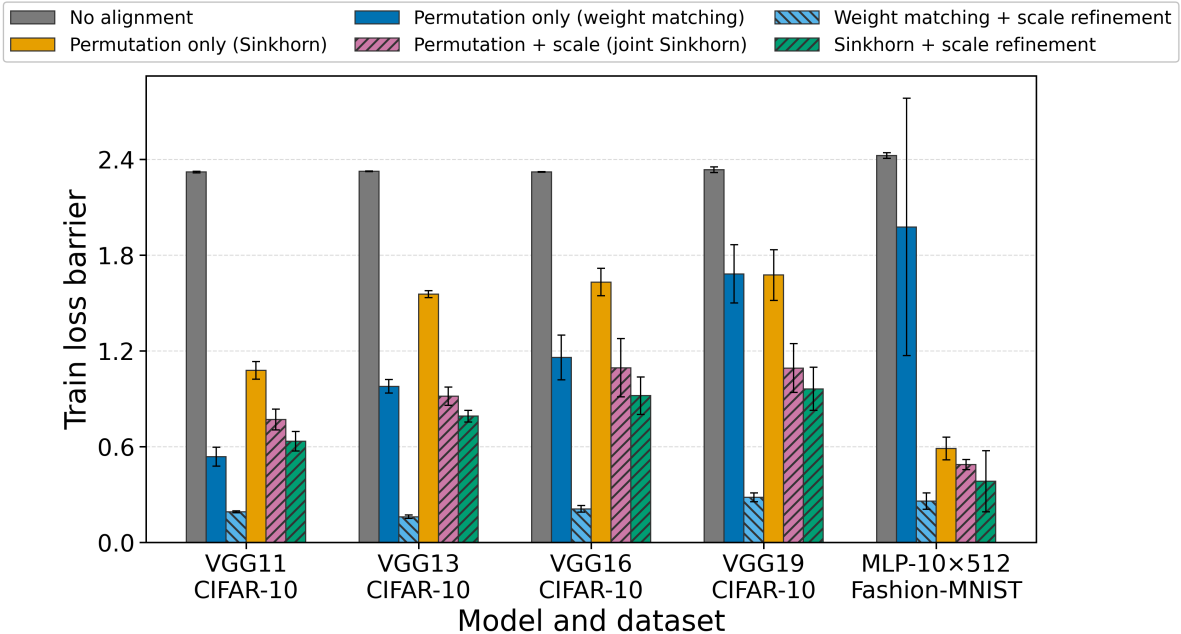


Figure 4: Training-loss barriers across the studied architecture–dataset combinations. Bars show the mean over three disjoint independently trained endpoint pairs, and error bars show the sample standard deviation. Hyperparameters are selected separately for each pair using validation data only.

6 DISCUSSION AND LIMITATIONS

Symmetry and redundancy play different roles.

Permutation removes discrete coordinate mismatch. Positive scaling differs: it is connected to the identity by a zero-loss exact-symmetry orbit and therefore cannot change whether an unrestricted continuous low-loss connection exists. In the nonlinear experiments, it can only make such a connection easier to express or find within the tested path families and optimization procedures. Width, by contrast, changes the architecture’s representational freedom and can create routes unavailable to a narrower network; curved paths relax the separate requirement that such a route be straight. These mechanisms are complementary but not interchangeable: selecting a better representative cannot create missing capacity, while additional capacity does not by itself select a favorable representative.

Implications for alignment studies. The exhaustive protocol is central to this interpretation. Without it, a high barrier after Sinkhorn or weight matching could always be ascribed to search failure. Conversely, the gap between joint Sinkhorn and fixed-permutation scale refinement warns against treating a failed continuous relaxation as evidence that a good functionally equivalent representative does not exist. Thus, the results do not reject a broader single-basin intuition; they show that its permutation-only formulation is too narrow, since positive scaling can lower barriers even

after permutation alignment. Moreover, weight and activation matching use parameter or representation similarity as proxies for linear connectivity, but these proxies need not select the lowest-barrier representatives. Sinkhorn directly optimizes a data-dependent interpolation objective, but remains sensitive to relaxation and optimization choices (Ainsworth et al., 2023; Peña et al., 2023). Better objectives will require identifying which properties of the endpoints and loss landscape determine linear compatibility.

Limitations and future work. The exact conclusions come from small one-hidden-layer networks trained on the XOR problem. The larger-network study is non-exhaustive, uses only three disjoint pairs per architecture, and relies on pair- and method-specific hyperparameter selection. Our nonlinear path searches can certify found paths but cannot certify nonexistence. The scale-aware soft transformation is exactly function preserving only after projection; during Sinkhorn optimization it is a relaxation. Finally, we consider ReLU networks, for which positive homogeneity holds, and do not address normalization layers, residual architectures, attention symmetries, or approximate functional equivalences. Larger multi-seed studies are needed before drawing broad architectural conclusions. A promising direction is to combine positive-scale refinement with activation repair (Jordan et al., 2023), which may yield still lower barriers.

7 CONCLUSION

A central problem in studying neural-network loss landscapes is determining whether barriers between independently trained solutions reflect genuine basin separation or removable parameterization mismatch. Permutation symmetry is an important but incomplete explanation of neural-network connectivity. Exhaustive XOR search shows that trained endpoints can remain incompatible under every permutation, for both minibatch SGD and full-batch GD. Positive scaling improves even the exhaustive best permutation, and it also lowers optimized nonlinear-path barriers at larger widths. However, scaling does not recover low-loss polygonal or Bézier connections for the difficult width-2 pairs. The width-dependent nonlinear experiment therefore shows that exact symmetries do not subsume representational constraints. The same training-loss benefit of scale-aware linear alignment transfers to fixed VGG/CIFAR-10 and MLP/Fashion-MNIST pairs. The central distinction is therefore simple: symmetry determines which representatives we compare; redundancy determines which paths are available.

AI Use Statement

Generative AI tools were used to edit and organize manuscript text, brainstorm presentation and interpretation choices, assist with literature searches, and support programming tasks including implementation, debugging, refactoring, plotting, and L^AT_EX editing. They were not used to generate experimental data or autonomously select the reported results. All AI-assisted text, code, analyses, citations, and interpretations were reviewed and verified by the authors. The authors made the final methodological and scientific decisions and take responsibility for all content and artifacts in this work.

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CHECKLIST

For each question, the selected answer is shown in brackets, followed where useful by a short justification.

- For all models and algorithms presented, check if you include:
 - A clear description of the mathematical setting, assumptions, algorithm, and/or model. [Yes; Sections 2–3 and Appendix A.]
 - An analysis of the properties and complexity (time, space, sample size) of any algorithm. [No; empirical behavior rather than asymptotic complexity is the focus.]
 - (Optional) Anonymized source code, with specification of all dependencies, including external libraries. [Yes; the supplemental material includes anonymized source code, a

- dependency lockfile, and vendored external libraries.]
2. For any theoretical claim, check if you include:
 - (a) Statements of the full set of assumptions of all theoretical results. [Not Applicable; no formal theorem is claimed.]
 - (b) Complete proofs of all theoretical results. [Not Applicable.]
 - (c) Clear explanations of any assumptions. [Yes; the positive-homogeneity assumption is stated in Section 3.]
 3. For all figures and tables that present empirical results, check if you include:
 - (a) The code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL). [Yes; the supplemental material includes the implementation, frozen configurations, public-dataset preparation, and reproduction instructions.]
 - (b) All the training details (e.g., data splits, hyperparameters, how they were chosen). [Yes; Appendix A.]
 - (c) A clear definition of the specific measure or statistics and error bars (e.g., with respect to the random seed after running experiments multiple times). [Yes; Eq. 1, Table 1, and the limitations in Section 5.]
 - (d) A description of the computing infrastructure used. (e.g., type of GPUs, internal cluster, or cloud provider). [Yes; the supplemental instructions give the GPU, CPU, memory, and scheduler requirements.]
 4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets, check if you include:
 - (a) Citations of the creator If your work uses existing assets. [Yes.]
 - (b) The license information of the assets, if applicable. [Yes; licenses and pinned revisions for vendored code are included in the supplement.]
 - (c) New assets either in the supplemental material or as a URL, if applicable. [Yes; anonymized source code is included in the supplemental material.]
 - (d) Information about consent from data providers/curators. [Not Applicable; only synthetic XOR and the public CIFAR-10 and Fashion-MNIST benchmarks are used.]
 5. If you used crowdsourcing or conducted research with human subjects, check if you include:
 - (e) Discussion of sensible content if applicable, e.g., personally identifiable information or offensive content. [Not Applicable.]
 - (a) The full text of instructions given to participants and screenshots. [Not Applicable.]
 - (b) Descriptions of potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable. [Not Applicable.]
 - (c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. [Not Applicable.]

The Role of Scaling Symmetries in Linear and Nonlinear Mode Connectivity of Neural Networks: Supplementary Materials

A REPRODUCIBILITY DETAILS

A.1 XOR training and exhaustive alignment

The XOR set is $\{(0, 0), (0, 1), (1, 0), (1, 1)\}$ with labels $\{0, 1, 1, 0\}$. Models have one ReLU hidden layer, biases, and one output logit. The primary endpoint protocol uses full-batch GD (implemented with the SGD optimizer) at learning rate 0.05 for at most 5000 epochs. `ReduceLROnPlateau` uses factor 0.5, patience 200, threshold 10^{-4} , and minimum learning rate 5×10^{-4} . Training stops when loss has not improved by 10^{-4} for 500 epochs. For each retained pair, all $h!$ permutations are evaluated. Interpolation selection prioritizes classification error in the width-2 diagnostic and loss barrier in the scale-aware study.

The width-2 robustness replication uses true minibatch SGD with minibatches containing two of the four XOR examples and reshuffles them every epoch. It otherwise uses the same initialization range, learning-rate schedule, stopping rule, maximum of 5000 epochs, random seeds, and exhaustive identity/swap evaluation. Eight successful endpoints produce 28 pairs in each protocol. Matching epoch counts does not match optimizer-update counts: the minibatch protocol makes two updates per epoch and is therefore a robustness comparison, not an isolated causal test of stochastic gradient noise. The width-3/5/7 scale-aware study uses the primary full-batch-GD endpoints.

For nonlinear paths, both quadratic Bézier curves and polygonal chains are optimized for 1500 steps at learning rate 0.05, with no weight decay and 31 path samples. The default path has one trainable interior point; harder pairs use up to ten.

A.2 Nonlinear path optimization with endpoint scaling

Let T_u denote the function-preserving positive-scaling action in Eq. 4, with all permutations set to the identity. We assign separate log-scales u_A and u_B to the two endpoints, producing the scaled representatives $T_{u_A}(\theta_A)$ and $T_{u_B}(\theta_B)$. For $q \in \{\text{poly}, \text{Bézier}\}$, let $\phi_v^q(\lambda; T_{u_A}(\theta_A), T_{u_B}(\theta_B))$ denote the corresponding path in Eqs. 7–8. The path control point and endpoint log-scales are optimized jointly using

$$\min_{v, u_A, u_B} \max_{\lambda \in \Lambda} \left[C(\phi_v^q(\lambda; T_{u_A}(\theta_A), T_{u_B}(\theta_B))) - ((1 - \lambda)C(\theta_A) + \lambda C(\theta_B)) \right]_+ + \lambda_s (\|u_A\|_2^2 + \|u_B\|_2^2), \quad (10)$$

where Λ is the set of sampled path locations and $[z]_+ = \max(z, 0)$. Because T_u preserves the represented function, the endpoint losses and therefore the baseline term in Eq. 10 remain unchanged. The first term minimizes the largest sampled loss excess, while the quadratic penalty discourages extreme scales.

The endpoint-scaling comparison uses one interior point, raw endpoints without permutation, and independently optimized positive scales at both endpoints. Path parameters and endpoint log-scales are optimized jointly with Adam for 1500 steps at learning rate 0.05; the scale penalty is 10^{-4} , and final barriers are evaluated on 501 path locations.

A.3 XOR scale-aware optimization

At each optimization step, XOR alignment evaluates binary cross-entropy at 31 evenly spaced interpolation points. The implementation subtracts the mean endpoint loss from each value and backpropagates the largest resulting excess; thus, the path-sample contribution to the gradient comes from the currently worst grid point rather than an average over all 31 locations. Final reported curves are evaluated on a separate 61-point grid.

The default permutation only (Sinkhorn) run uses Adam for 500 steps, learning rate 0.01, temperature 1, 20 Sinkhorn iterations, and identity strength 1. Scale refinement uses Adam for 500 steps, learning rate 0.01, and log-scale regularization 10^{-3} . Joint optimization uses the same default budget. If a default run misses its target, a bounded search over learning rates, temperatures, and regularization is used; settings are selected per pair. The scale target is barrier $\epsilon = 0.01$, while permutation-only Sinkhorn targets that pair’s exhaustive barrier.

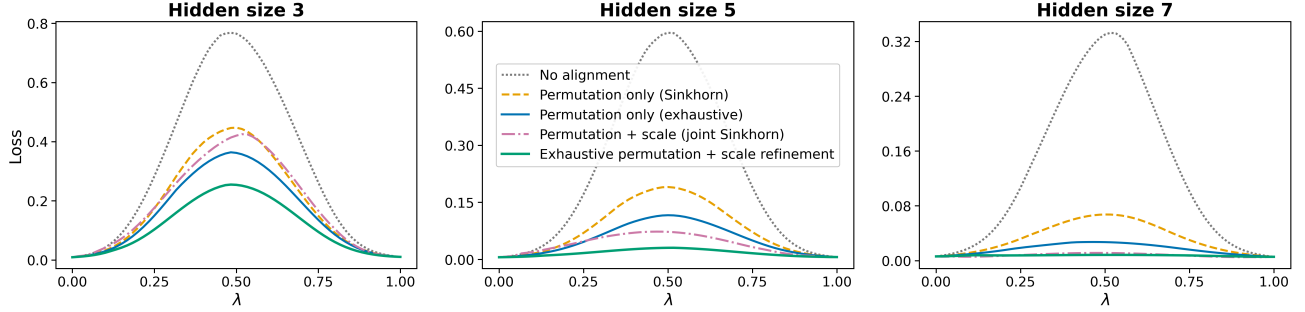


Figure A.1: Mean XOR interpolation-loss profiles at different hidden widths, comparing four alignment variants with raw interpolation. At each λ , loss is averaged across endpoint pairs.

A.4 Larger-network experiments

The VGG architectures and endpoint-training code are adapted from the `pytorch-vgg-cifar10` implementation of Fu (2017). We modified the training code for deterministic independent runs and checkpoint generation, and trained all endpoints from scratch; no pretrained upstream checkpoints are used.

For every architecture, Figure 4 uses the final-final cell for three disjoint run pairs, (0, 1), (2, 3), and (4, 5), where each integer is the random seed of a complete independent training run. Weight matching is computed for each pair, and every data-dependent alignment or scale-refinement method is fitted separately.

Endpoint models are trained only on the endpoint-training split. For each optimized method and endpoint pair, hyperparameter candidates are fitted and early-stopped on the tuning half of the validation set, then compared on the disjoint selection half. The selected configuration is refit at full budget on the tuning split. Training and test barriers are evaluated only after the hyperparameters and alignment artifacts are frozen; the official test set is used solely for final evaluation.

The permutation-then-scale methods optimize only log-scales after fixing the corresponding hard permutation. All six methods are evaluated on the same 61-point interpolation grid. Figure 4 reports the mean and sample standard deviation across the three independent pairs on the fixed training-report subset; Figure A.2 reports the corresponding official-test-set result.

The validation grids are listed in Table 2. Selection is performed independently for every endpoint pair, so the chosen tuple can differ across pairs.

Method	η	τ	λ_s
Weight matching + scale refinement	{0.005, 0.02, 0.08}	–	$\{10^{-4}, 10^{-3}, 3 \times 10^{-3}, 10^{-2}\}$
Permutation only (Sinkhorn)	{0.03, 0.10, 0.30, 0.75}	{1, 2.5, 5}	–
Permutation + scale (joint Sinkhorn)	{0.01, 0.05, 0.20}	{1, 2.5}	$\{3 \times 10^{-4}, 3 \times 10^{-3}\}$
Sinkhorn + scale refinement	{0.005, 0.02, 0.08}	–	$\{10^{-4}, 10^{-3}, 3 \times 10^{-3}, 10^{-2}\}$

Table 2: Hyperparameter grids for the larger-network alignment methods. Here η is the optimizer learning rate, τ is the Sinkhorn temperature, and λ_s is the coefficient of the quadratic log-scale penalty. A dash denotes a parameter that does not apply. All applicable combinations are evaluated separately for each endpoint pair.

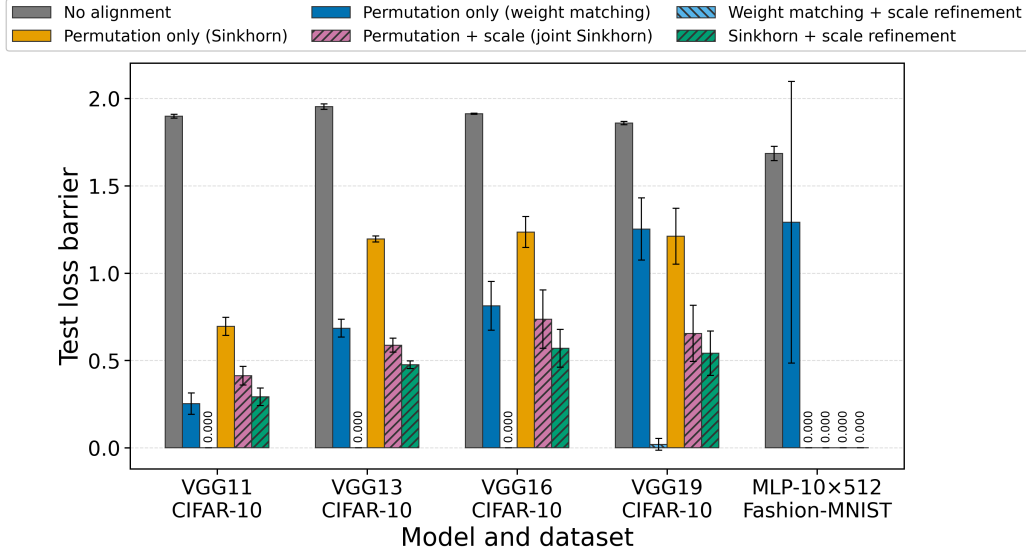


Figure A.2: Test-loss barrier for the same frozen alignments and three endpoint pairs as Figure 4. Bars show the mean and error bars show the sample standard deviation. The complete official test set is accessed only after hyperparameters and alignment artifacts are frozen.

Tables 3 and 4 give the exact aggregates underlying the training and test bar plots, respectively.

Method	VGG11 CIFAR-10	VGG13 CIFAR-10	VGG16 CIFAR-10	VGG19 CIFAR-10	MLP Fashion-MNIST
No alignment	2.320 ± 0.005	2.325 ± 0.002	2.320 ± 0.002	2.334 ± 0.018	2.423 ± 0.019
Permutation only (weight matching)	0.536 ± 0.059	0.976 ± 0.042	1.157 ± 0.141	1.681 ± 0.183	1.976 ± 0.807
Weight matching + scale refinement	0.191 ± 0.006	0.159 ± 0.011	0.209 ± 0.021	0.281 ± 0.028	0.258 ± 0.051
Permutation only (Sinkhorn)	1.077 ± 0.056	1.555 ± 0.021	1.630 ± 0.087	1.675 ± 0.159	0.587 ± 0.071
Permutation + scale (joint Sinkhorn)	0.769 ± 0.064	0.915 ± 0.057	1.093 ± 0.183	1.092 ± 0.153	0.487 ± 0.031
Sinkhorn + scale refinement	0.633 ± 0.061	0.790 ± 0.036	0.918 ± 0.117	0.961 ± 0.135	0.382 ± 0.191

Table 3: Train-loss barrier by architecture and alignment method, reported as mean $\pm$ sample standard deviation over three disjoint endpoint pairs. These values underlie Figure 4.

Method	VGG11 CIFAR-10	VGG13 CIFAR-10	VGG16 CIFAR-10	VGG19 CIFAR-10	MLP Fashion-MNIST
No alignment	1.899 ± 0.010	1.953 ± 0.015	1.913 ± 0.003	1.859 ± 0.010	1.685 ± 0.041
Permutation only (weight matching)	0.253 ± 0.061	0.684 ± 0.051	0.813 ± 0.139	1.252 ± 0.178	1.291 ± 0.806
Weight matching + scale refinement	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.019 ± 0.034	0.000 ± 0.000
Permutation only (Sinkhorn)	0.695 ± 0.052	1.196 ± 0.017	1.235 ± 0.089	1.211 ± 0.160	0.000 ± 0.000
Permutation + scale (joint Sinkhorn)	0.412 ± 0.053	0.588 ± 0.039	0.736 ± 0.167	0.655 ± 0.161	0.000 ± 0.000
Sinkhorn + scale refinement	0.292 ± 0.051	0.475 ± 0.022	0.570 ± 0.109	0.542 ± 0.127	0.000 ± 0.000

Table 4: Test-loss barrier by architecture and alignment method, reported as mean $\pm$ sample standard deviation over the same three endpoint pairs as Table 3. These values underlie Figure A.2.

Test-loss barriers are generally smaller than their training-loss counterparts. The barrier is measured relative to the linear interpolation of the endpoint losses, not to zero loss. In the Fashion-MNIST pairs, for example, endpoint training losses are 0.020–0.041, whereas endpoint test losses are 0.681–0.809; an interior test-loss curve can therefore remain below this much higher baseline even while the same path rises far above the near-zero training baseline. Consequently, the zero sampled Fashion-MNIST test-loss barriers in Figure A.2 do not imply zero training-loss barrier or constant classification error along the path.